

Sequence diagram

Behavioral Modeling

- make a list of the different states of a system (How does the system behave?)
- indicate how the system makes a transition from one state to another (How does the system change state?)
 - indicate event
 - indicate action
- draw a **state diagram** or a **sequence diagram**

Sequence Diagram

- From the flow of events in the use case or scenario proceed to the sequence diagram
- A sequence diagram is a graphical description of objects participating in a use case or scenario using a DAG (direct acyclic graph) notation
- Relation to object identification:
 - Objects/classes have already been identified during object modeling
 - Objects are identified as a result of dynamic modeling

Sequence Diagram (cont'd)

- Heuristic:
 - An event always has a sender and a receiver.
 - The representation of the event is sometimes called a message
 - Find them for each event => These are the objects participating in the use case

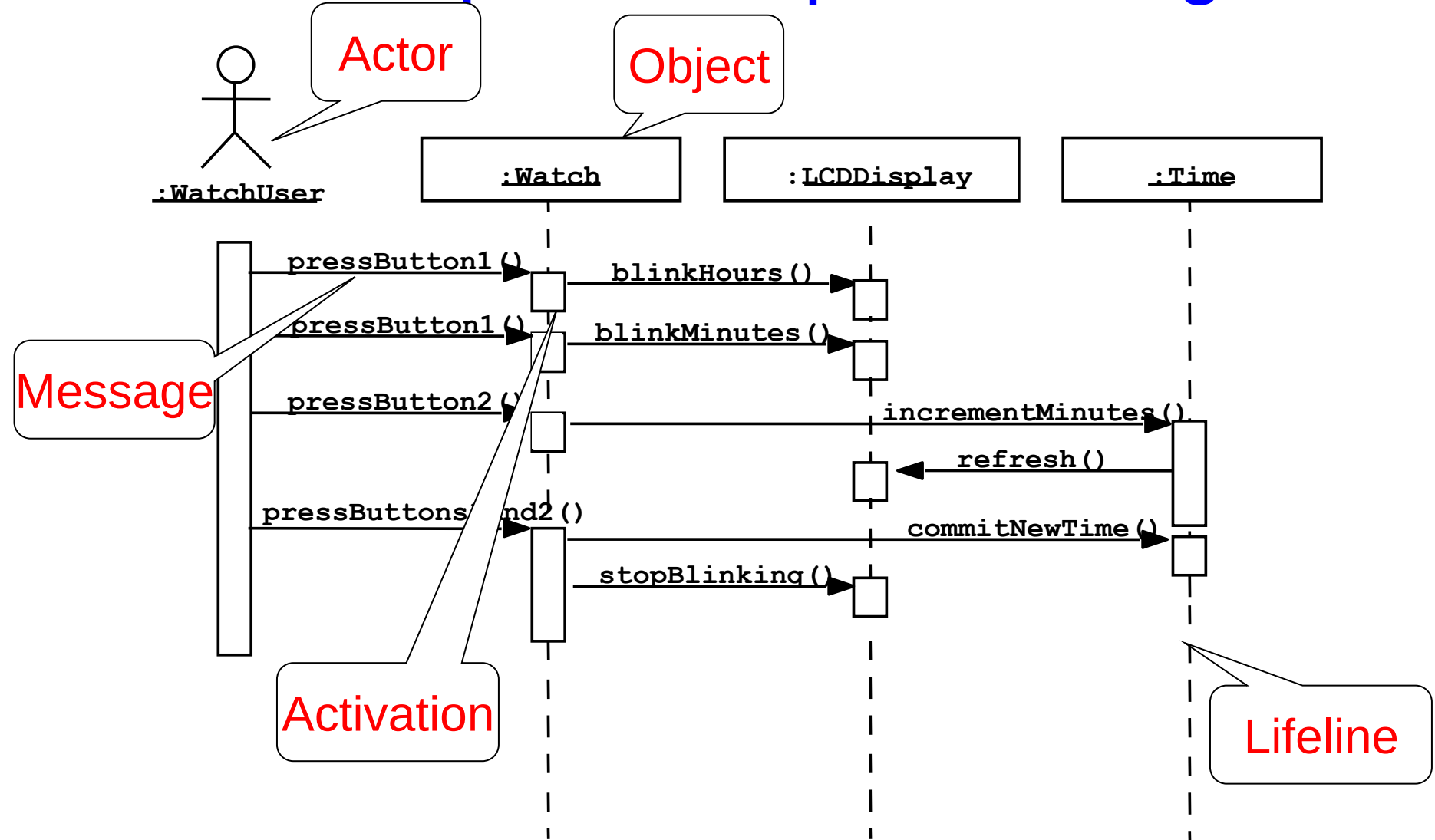
Heuristics for Sequence Diagrams

- Layout:
 - 1st column: Should correspond to the actor who initiated the use case
 - 2nd column: Should be a boundary object
 - 3rd column: Should be the control object that manages the rest of the use case
- Creation:
 - Control objects are created at the initiation of a use case
 - Boundary objects are created by control objects

Heuristics for Sequence Diagrams (cont'd)

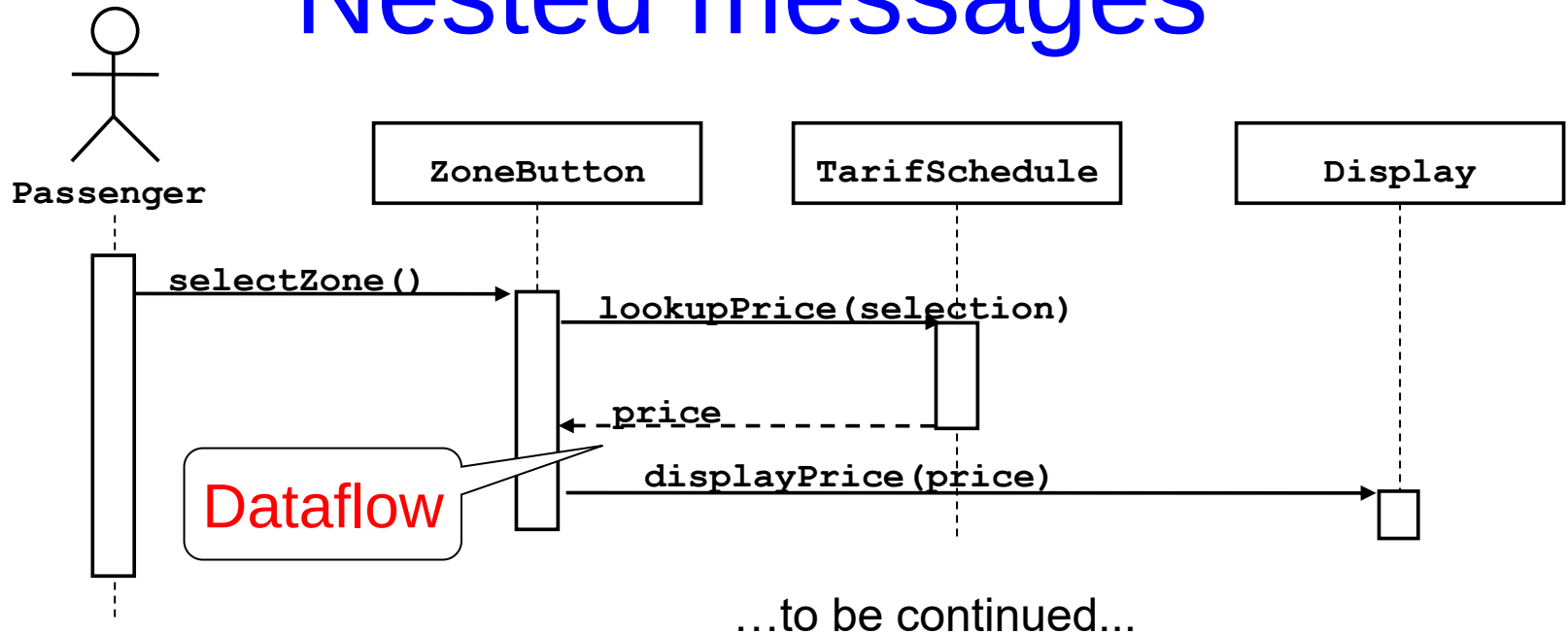
- Access:
 - Entity objects are accessed by control and boundary objects,
 - Entity objects should never call boundary or control objects: This makes it easier to share entity objects across use cases and makes entity objects resilient against technology-induced changes in boundary objects.

UML first pass: Sequence diagram



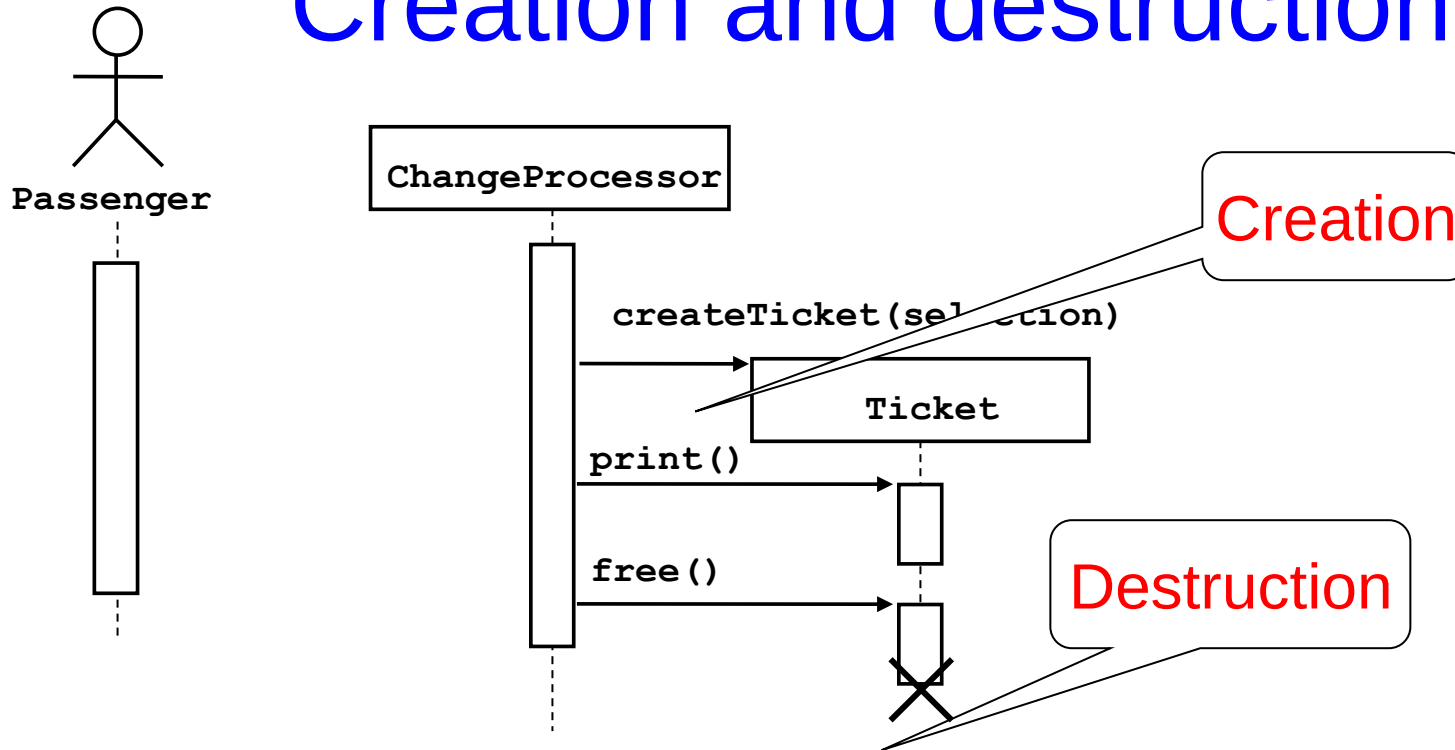
Sequence diagrams represent the behavior as interactions⁷

Nested messages



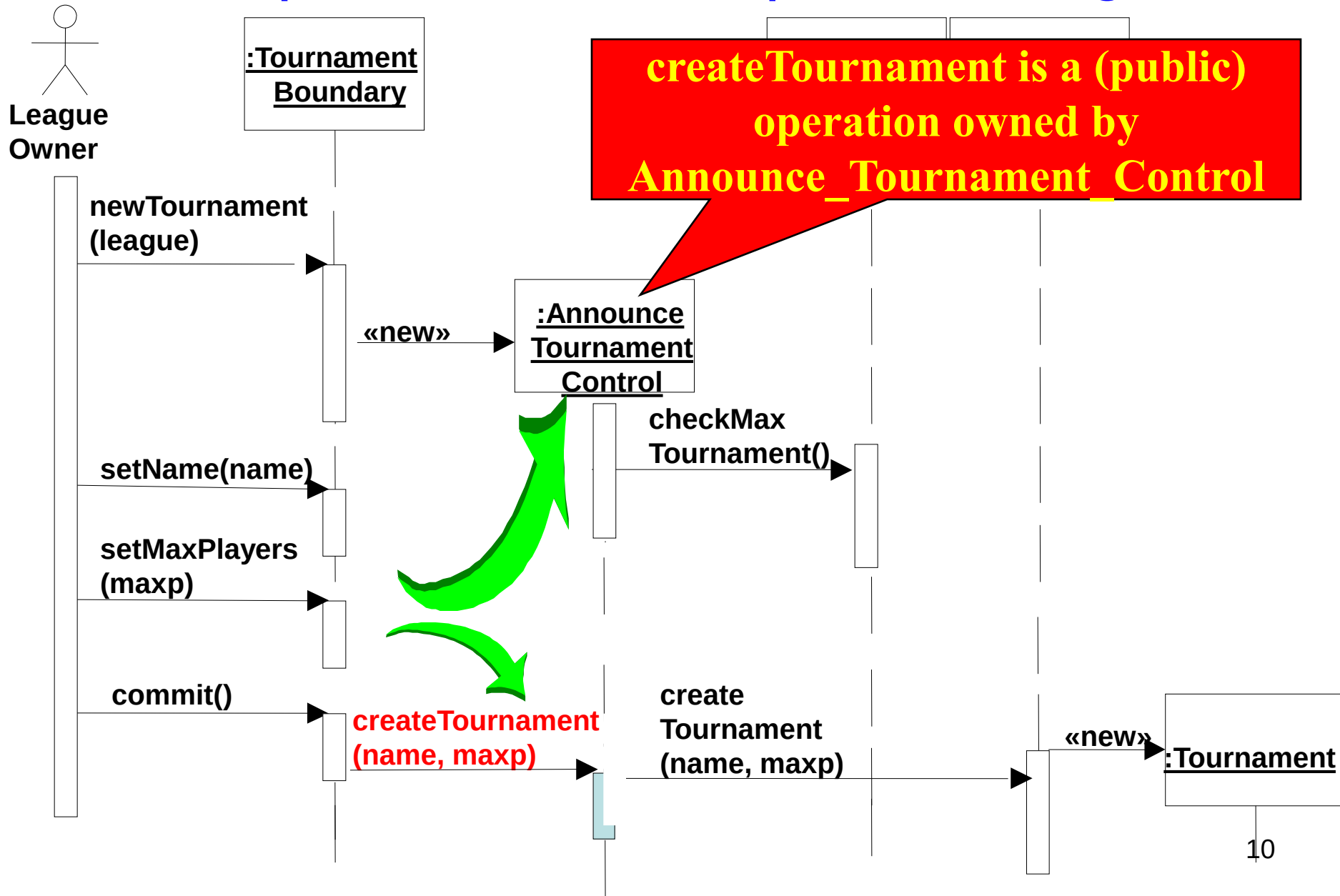
- The source of an arrow indicates the activation which sent the message
- An activation is as long as all nested activations
- Horizontal dashed arrows indicate data flow
- Vertical dashed lines indicate lifelines

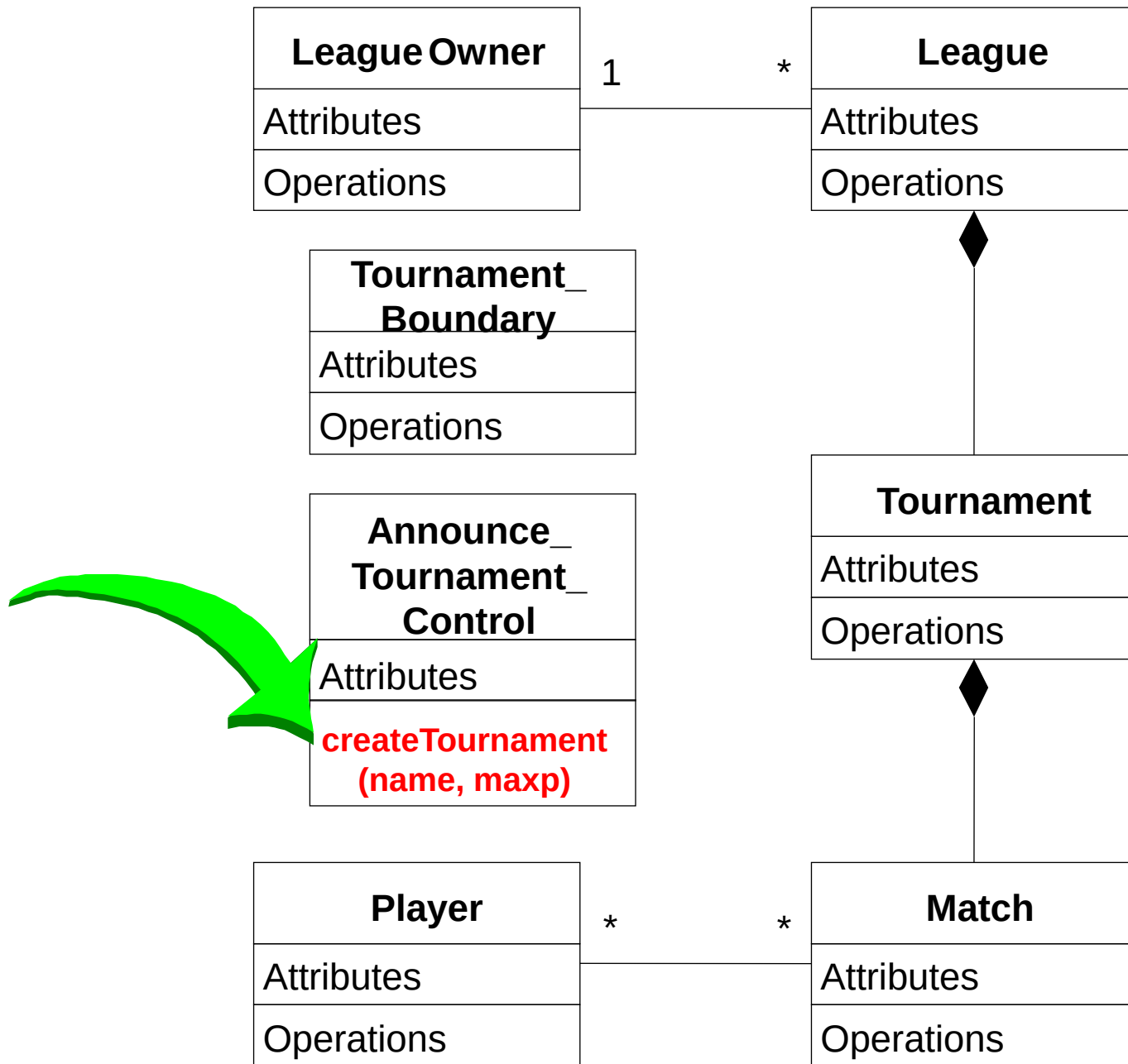
Creation and destruction



- Creation is denoted by a message arrow pointing to the object.
- Destruction is denoted by an X mark at the end of the destruction activation.
- In garbage collection environments, destruction can be used to denote the end of the useful life of an object.

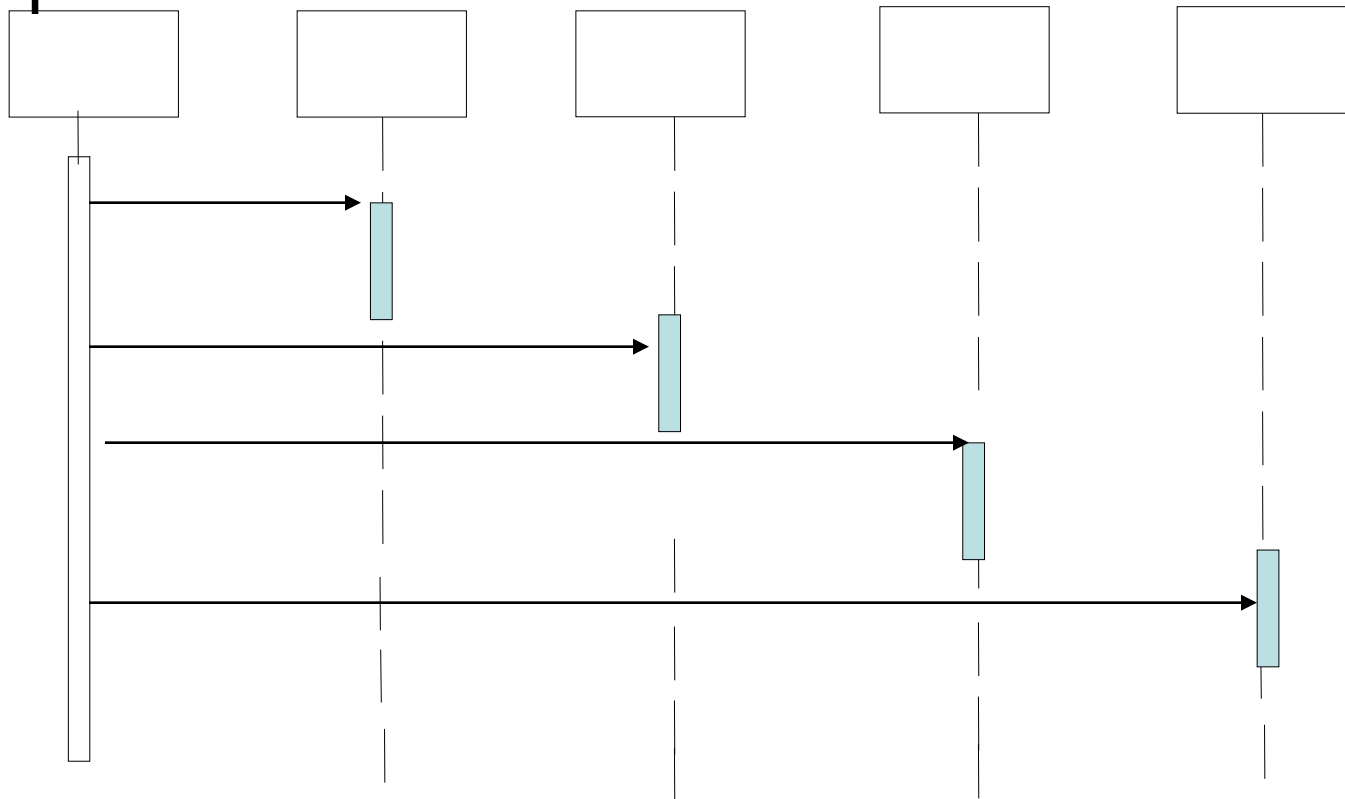
Example from the Sequence Diagram





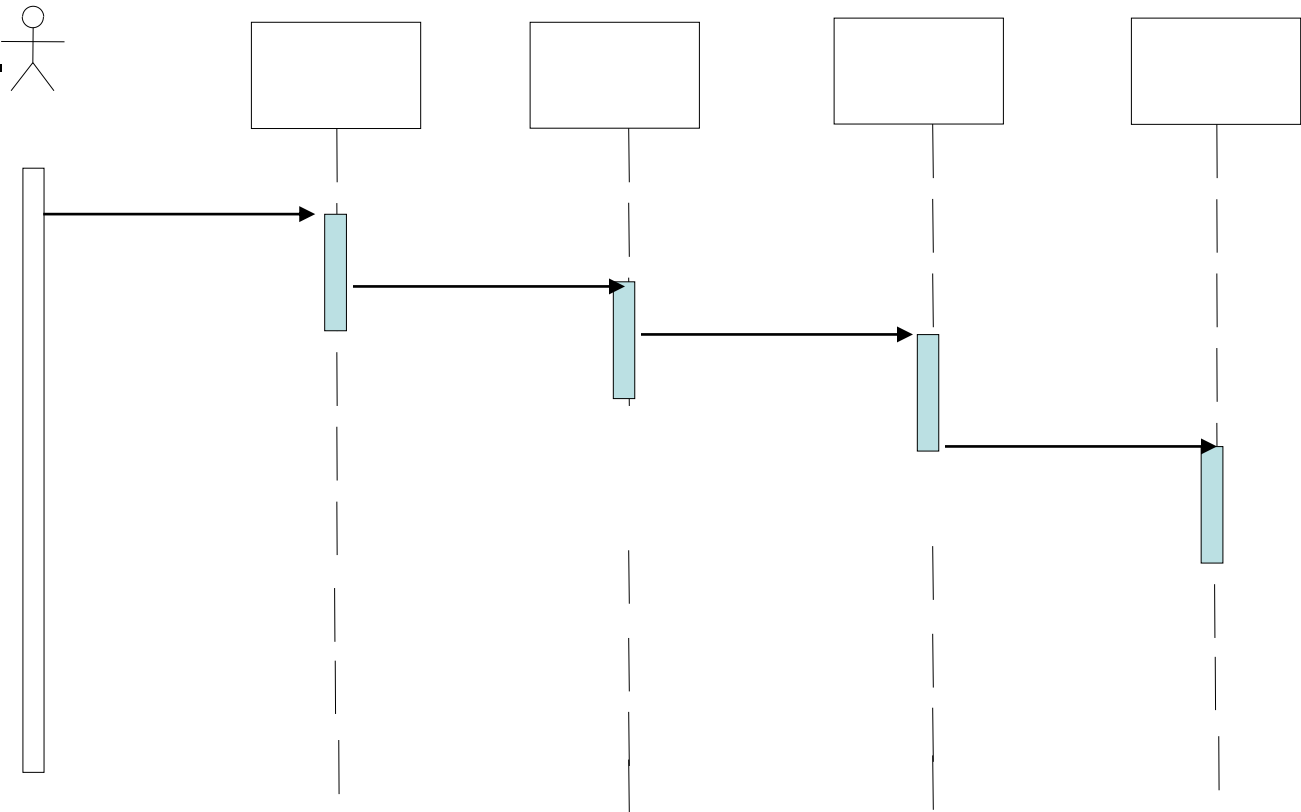
Fork Diagram

- Much of the dynamic behavior is placed in a single object, usually the control object. It knows all the other objects and often uses them for direct questions and commands.

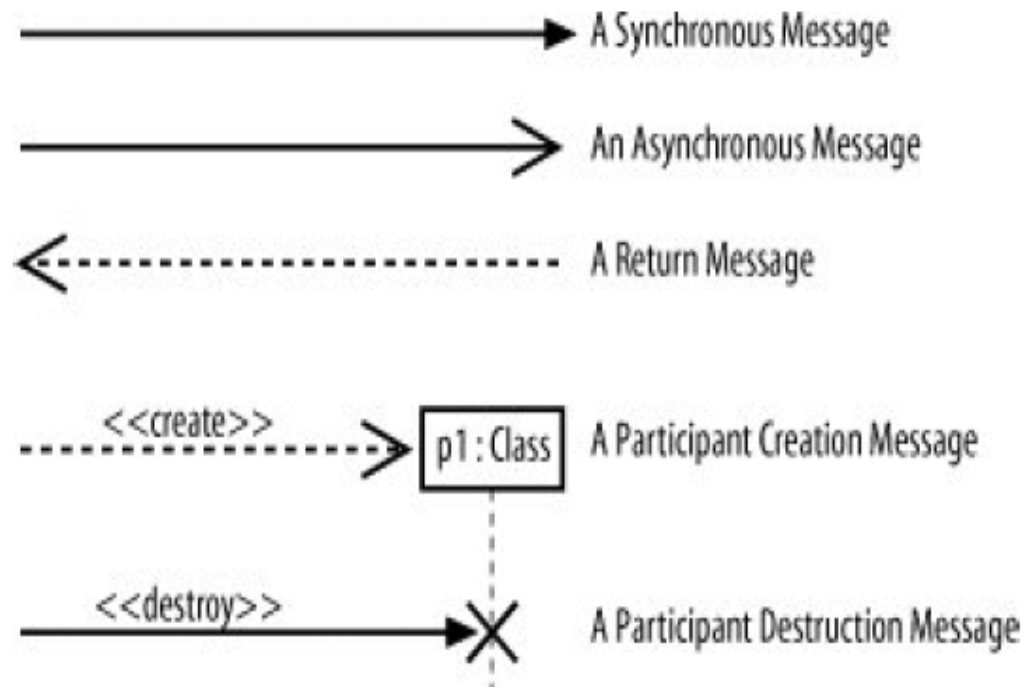


Stair Diagram

- The dynamic behavior is distributed. Each object delegates some responsibility to other objects. Each object knows only a few of the other objects and knows which objects can help with a specific behavior.



Sequence Diagram



Q&A